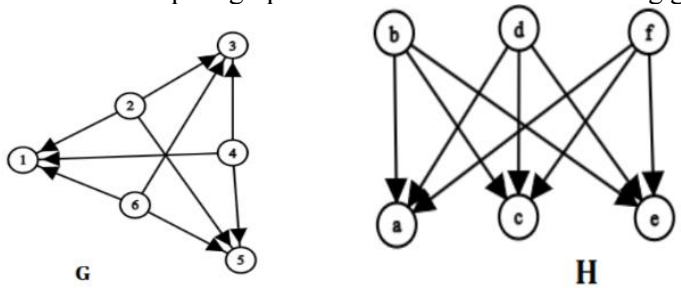
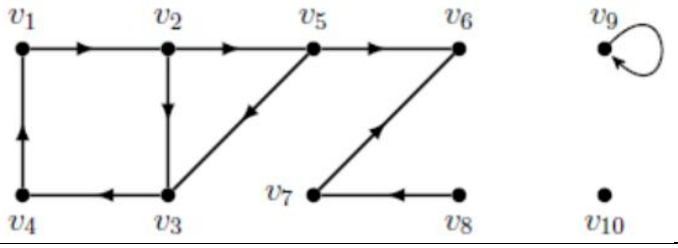
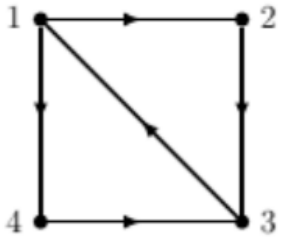
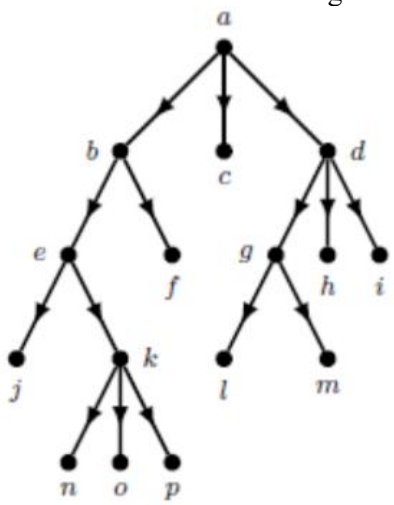
DMGT (BE04000261)

QUESTIONS FOR VIVA

SEMESTER 4 – CE

1	A committee of 5 persons, is to be formed from 6 men and 4 women. In how many ways this can be done when (i) at least 2 women are included, (ii) at most 2 women are included.
2	Solve the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 2$ with initial condition $a_0 = 1$ and $a_1 = -1$ using method of undetermined coefficients.
3	If $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and operation $+_8$ is an addition modulo 8, then show that $(G, +_8)$ is an abelian group.
4	Let Q_+ be the set of all positive rational numbers and $*$ be a binary operation on Q_+ defined by $a * b = ab/3$ then show that $(Q_+, *)$ is an abelian group.
5	Let A be a given finite set and $\rho(A)$ its power set. Let $\subseteq$ be the inclusion relation on the elements of $\rho(A)$. Draw the Hasse diagram of $\langle \rho(A), \subseteq \rangle$ for $A = \{a, b, c, d\}$.
6	Draw Hasse diagram for $P = \{1, 2, 3, 5, 11, 13, 17, 19, 23\}$ and $\leq$ is a relation such that $x \leq y \Leftrightarrow x \mid y$.
7	Check whether the relation R defined in the $\{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b) : b = a + 1\}$ is reflexive, symmetric or transitive.
8	Let $X = \{1, 2, 3, \ldots, 7\}$ and $R = \{\langle x, y \rangle \mid x - y$ is divisible by $3\}$. Show that R is an equivalence relation on X. Draw the graph of R.
9	Let $A = \{1, 2, 3, 4\}$. For the relation R whose matrix is given, find the matrix of the transitive closure by using Warshall's algorithm. $$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
10	Draw the following graphs: (i) $K_{3,4}$ (ii) $K_{4,4}$ (iii) $K_{5,4}$ (iv) K_5
11	Define Isomorphic graphs. Check whether the following graphs are isomorphic. 
12	Find reachable set of each node of the given digraph.

	
13	Define path matrix. Use Warshall's algorithm to obtain path matrix from the adjacency matrix of following graph 
14	In which order does a pre-order, in-order and post-order traversal visit the vertices of the ordered rooted tree shown in figure? 
15	Use a truth table to determine whether the following argument form is valid. $$p \rightarrow q$$ $$q \rightarrow r$$ $$\therefore p \rightarrow r$$